

Healthcare Claims Cost Analysis (SQL)

Executive Summary | Built in Microsoft SQL Server / SSMS

Last updated: January 14, 2026

Project goal

Analyze healthcare claims to identify cost drivers and reimbursement patterns, including top claim types, procedure/diagnosis codes, high-cost members, and billed-vs-paid ratios to surface areas for cost containment and contract review.

Data and tools

- Source: Claims flat file imported into SQL Server (table: dbo.claims).
- Key fields: claim_type, billed_amount, paid_amount, cpt_code, icd_code, member_id, provider_id.
- Tools: Microsoft SQL Server, SQL Server Management Studio (SSMS).
- Privacy: [TODO - confirm dataset is synthetic/de-identified and note any PHI handling].

Core analyses delivered

- Claim type cost breakdown: totals and ranking by paid amount.
- Top CPT and ICD cost drivers: total paid and average paid per claim.
- Member-level spend: top members and claim-type drivers of cost.
- Billed vs paid ratios by claim_type, CPT, and provider (weighted by dollars).

Key findings (replace placeholders with your results)

- Top claim type by total paid: [TODO claim_type] (\$[TODO]).
- Largest CPT driver: [TODO CPT] (\$[TODO]) with avg paid/claim \$[TODO].
- Largest ICD driver: [TODO ICD] (\$[TODO]).
- High-cost members are primarily driven by: [TODO inpatient/outpatient/ER/pharmacy mix].
- Lowest paid ratio claim type(s): [TODO] (paid-to-billed ~[TODO]).
- Providers/CPTs with paid ratio > 1.0 flagged for review (possible adjustments/COB/data quirks).

Business implications

- Targeted cost containment: focus on the highest-spend claim types and top CPT drivers.
- Contracting and reimbursement: investigate low paid-ratio categories for negotiation opportunities.
- Care management: high-cost members can be prioritized for outreach and utilization review.
- Data quality controls: monitor outlier paid ratios (very low/high) and negative adjustments.

Next steps with more time

- Add time dimension (month/quarter) to identify trend and seasonality.
- Segment by plan/product line and network status (in/out-of-network).
- Incorporate allowed amount and adjustment types for cleaner reimbursement analysis.
- Create a lightweight dashboard in Power

Repository contents and how to run

Included: README.md (context + instructions), sql/analysis.sql (all queries), outputs/*.csv (exported result tables), images/ (optional screenshots).

To reproduce: import the claims file into dbo.claims, then execute sql/analysis.sql in SSMS. Export key result sets to CSV for sharing.